

SmartWatt

Smart Energy Forecasting and Management System

GRADUATION PROJECT – 2026



Introduction

- ✓ Increasing electricity demand and limited awareness lead to inefficient energy use and higher costs.
- ✓ SmartWatt forecasts electricity consumption and provides actionable recommendations.
- ✓ It aims to reduce energy waste, support data-driven decisions, and contribute to Saudi Vision 2030 sustainability goals.

METHODOLOGY

- Data preprocessing (cleaning, datetime parsing, resampling to hourly, and outlier removal)
- Time-series preparation and train/test split
- Forecasting using ARIMA (2,1,2) based on historical consumption data
- Model evaluation using MAE and RMSE
- Rule-based recommendation engine for alert classification
- Visualization through an interactive Streamlit dashboard

CONCLUSION

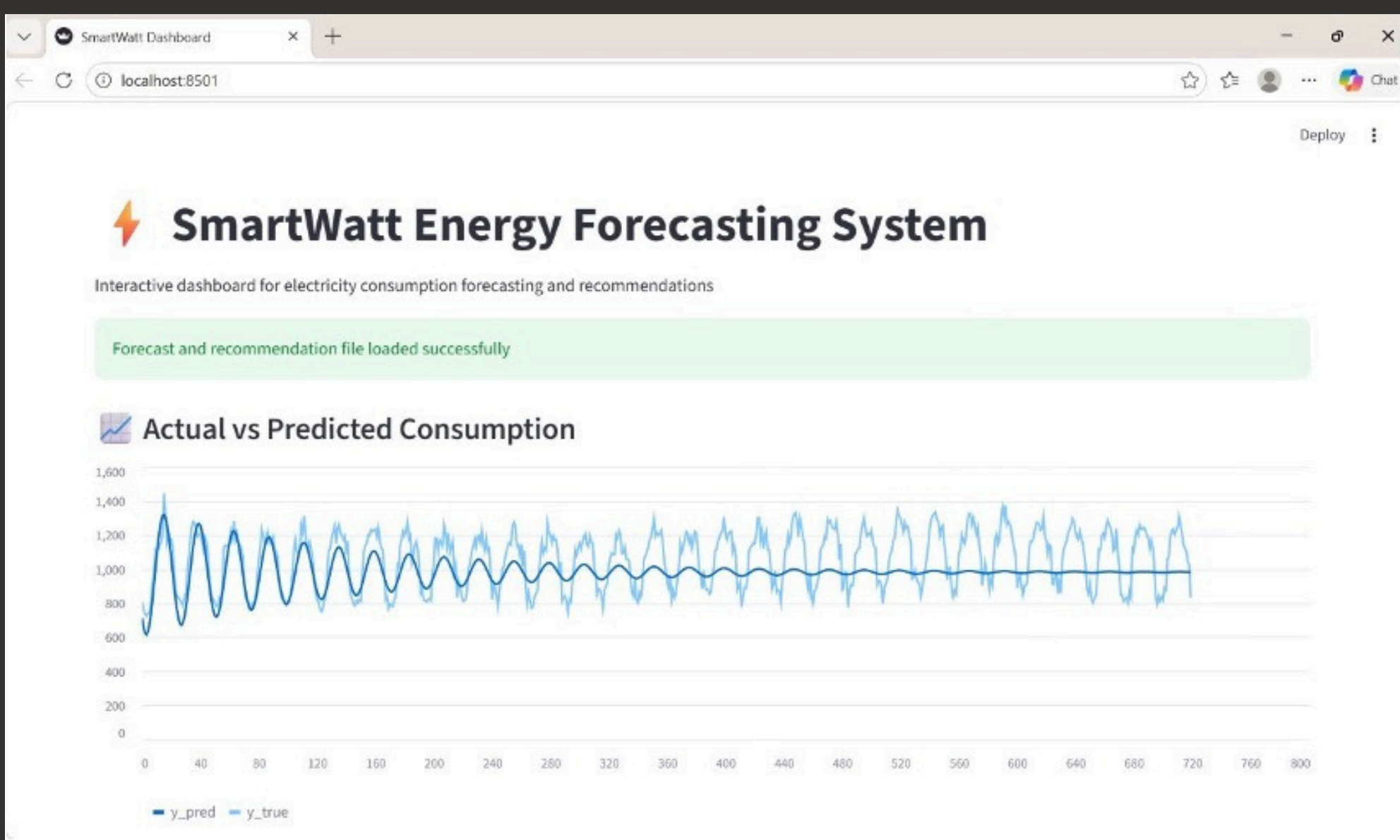
- Accurate forecasting using ARIMA
- Reliable performance (MAE: 126.81, RMSE: 150.54)
- Provides actionable energy-saving recommendations
- Enhances user awareness and reduces electricity costs
- Supports sustainability and Saudi Vision 2030

EXPERIMENTS

- Experimental Setup :**
- Historical electricity dataset
 - ARIMA (2,1,2) model
 - Python & Streamlit
 - Evaluation using MAE and RMSE

- Evaluation Metrics :**
- MAE = 126.81
 - RMSE = 150.54

- Results :**
- The model captured the general consumption pattern
 - Predictions closely follow actual consumption trends
 - The model shows stable and reliable performance



Actual vs Predicted Electricity Consumption

PROJECT SUPERVISOR

Dr. Sulaiman Alamro

TEAM MEMBERS

Raydaa ALluhaydan - 431201378

Aseel Alruthea - 431201848

Wasayf Alanazi - 422205666

Budur Alanazi - 421214618

Rawan Aloufi - 431201747